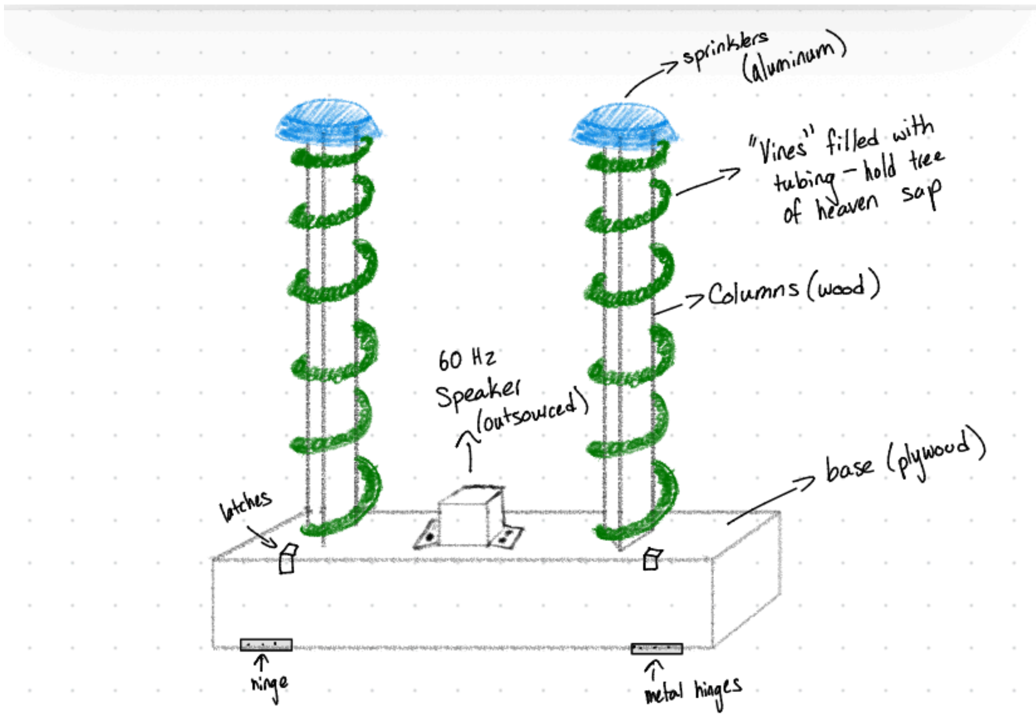
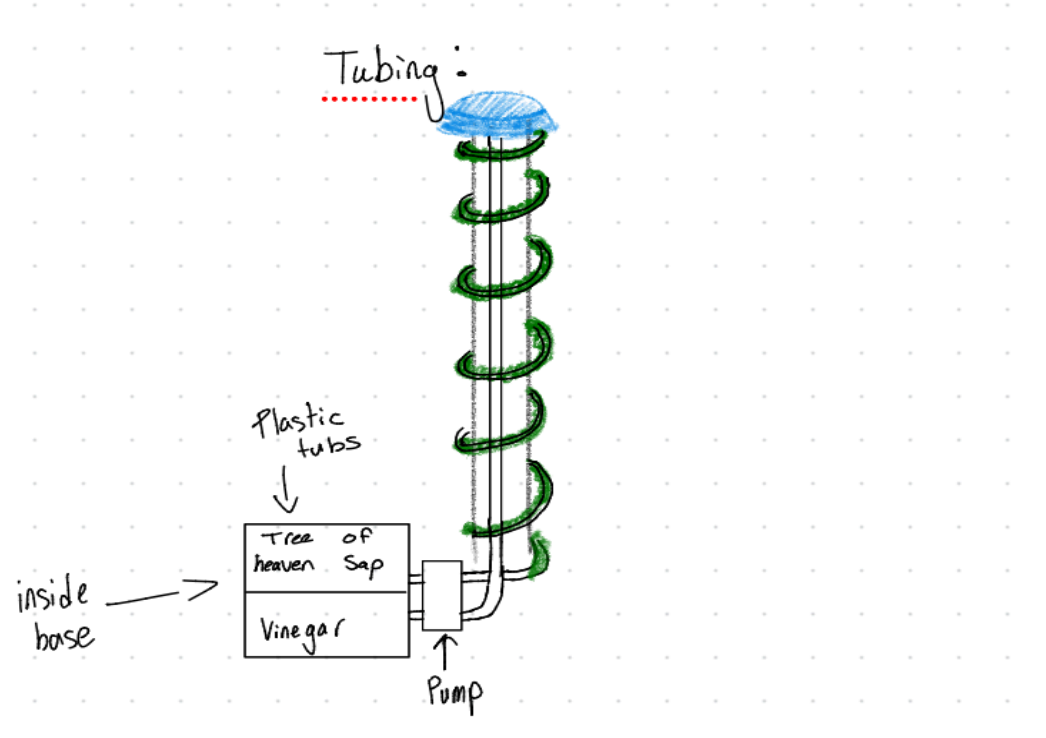
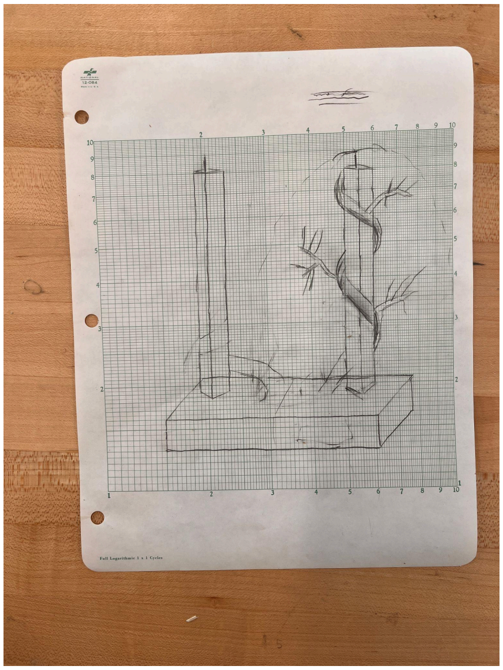
ODP 4: Planning Prototype

Initial Sketch:





Mock-Up Prototype:



Timeline:

Week of March 2nd:

- [x] ~~Begin first sketches~~
- [x] ~~We planned to get the first prototype done by the end of lab for the week of March 2nd which was completed~~

Week of March 9th:

- [ ] Finalized sketch including all logistics
    - dimension, exact tubing system set up, finalized design
- [ ] Begin working on CAD for second prototype
- [ ] Have all parts ordered for second prototype

Week of March 16th:

- [ ] Have finalized CAD for second prototype
- [ ] Begin working on 2nd prototype

Week of March 23rd:

- [ ] Continue working on 2nd prototype
- [ ] Order any parts needed for final prototype

SPRINK BREAK (No Lab)

Week of April 6th:

- [ ] Test second prototype
- [ ] Change CAD accordingly to finalized prototype
    - [ ] Functionality, Make sure all pumps and sprinklers work
- [ ] Begin working on final prototype

Week of April 13th:

- [ ] Continue working on final prototype

Week of April 20th:

- [ ] Test Final Prototype accordingly and finalize product

Questions:

1. Will the artificial grapevine prototype be able to adequately mimic a real grapevine in order to draw in the lantern flies while maintaining the aesthetics of the vineyard?

- The grapevine will be able to adequately mimic a real grapevine, as there are multiple design parameters we added to the prototype that allowed us to visualize how similar it would look. Granted, our prototype was made out of cardboard and tissue paper but it helped us see whether making a similar model to actual vineyards would be possible.

2. What will the general layout of the box be and how will the various components be mounted and fit together (speaker, sap reservoir, etc.)?

- As seen in the prototype, the box will be the support mount to two wooden poles that will be mounted on top. These poles will be the base for our tubing that will carry both the artificial tree of heaven sap, and the substance used to kill the spotted lantern flies. The tubing will be wrapped around each of the two poles mimicking the way grape vines wrap around trulleses. Then using a needle there will be numerous tiny holes poked through the tubing so the sap can leak out just a bit.
- The speaker will be mounted between the poles inside the box. Since it will be placed on the base, the sound will most likely also vibrate the structure, which will further attract the SFLs.

3. How many adult spotted lantern flies will be able to fit on the prototype? How will this compare to the final scaled product?

- Due to the first prototype, we definitely got a better idea of the size we want to make for this class, which will most likely be able to fit 80-100 SFL per tower so 200 total. This was tested by cutting a paper of similar dimensions to SFLs and comparing them to the towers.
- We also decided that the final prototype will most likely be a little smaller than this first one in order to stay within the budget and utilize storage space well.

Design Estimates:

Starting with power estimates, the two powered features we plan on integrating into our system are a fluid pump to pump tree of heaven sap as well as the liquid we will use to kill the flies and a speaker to emit 60Hz sound waves. When the sap is secreted from the tubes in the vines, we need a very slow drip. We envision a very light coating of the sap on the tubes so we can prolong usage without having to refill the system. In this case, some kind of peristaltic pump should be used for such a low flow rate.

Making some basic assumptions of what the flow rate may look like, we wish to assume that the system uses 4mm diameter tubing with ~30 holes uniformly spaced throughout the tube. Going with liquid coming out of the holes at around 1drop/min i.e. 0.05mL/min since a drop of water is around 0.05mL, we get that we need a flow rate of 1.5mL/min or around 90mL/hr. Looking up some basic pumps, micro peristaltic pumps are what output this required flowrate. The assumption of 1drop/min is also quite large so the pump can often be turned on and off periodically to fine tune the desired flow rate. The power draw of a micro 12V peristaltic pump is around 2-5W

For the speaker, a 1.75" subwoofer can deliver the desired 60 Hz sound at 72 decibels, meaning the sound will travel well over 500 m. This speaker uses 10 watts of power. The speaker would only be turned on during times the spotted lantern flies are active, meaning its power draw can be regulated

In order to power a more advanced prototype for a couple of hours, we think that a 12V ~5000mAh 3S lipo battery should suffice with some kind of Power Distribution Panel that is able to power the desired 12V modules we have in our system

When it comes to the size of our next prototype, it will be slightly bigger than our mockup. The base that held the trellis of the mockup will certainly be bigger because it needs to store the electronics and sap. Most of the exterior can be made from plywood. We are focusing primarily on functionality for this prototype rather than aesthetics. This means we want to get the fluid system, electronics, and structure much more fleshed out.

Fabrication and Assembly

Structural Components	Fluid System Components	Electrical System Components
Fabricated		
• 2 vertical columns (wood dowels or square wood posts) • Plywood boards • 3D printed vine tubing supports • 3D printed sprinkler cap (top dome) • Internal funnel/capture chamber • Speaker mounting bracket • Reservoir housing insert • Base internal divider walls • 3D printed tubs for holding vinegar and sap	• Tubing routing clips • Reservoir mounting bracket	• Battery mounting tray • Wire strain relief clips • Switch mounting panel
Purchased		
• Screws • Hinges • Latches • Fasteners (screws, bolts, nuts) • Silicone sealant	• 12V micro peristaltic pump (2–5W) • 4mm ID plastic tubing • Tubing Connectors • 250–500mL sealed reservoir container • Vinegar • Artificial Tree of Heaven sap mixture	• 12V battery (5000mAh) • 12W vibration speaker • Cycle timer • On/off switch • Wiring • Heat shrink tubing • Terminal connectors

2. Assembly Process

Step 1 – Base Construction

- **Assemble plywood base box by cutting sides and screwing together**
- **Install hinges and latch**
- **Install internal mounts for electrical components**
- **Secure two wooden columns to base using screws**

Step 2 – Fluid System Installation

- **Mount fluid reservoirs inside base**
- **Mount respective peristaltic pumps**
- **Connect:**
  Reservoir → Pump → Tubing → Vine loops
- **Add check valve**
- **Pressure test system with water.**

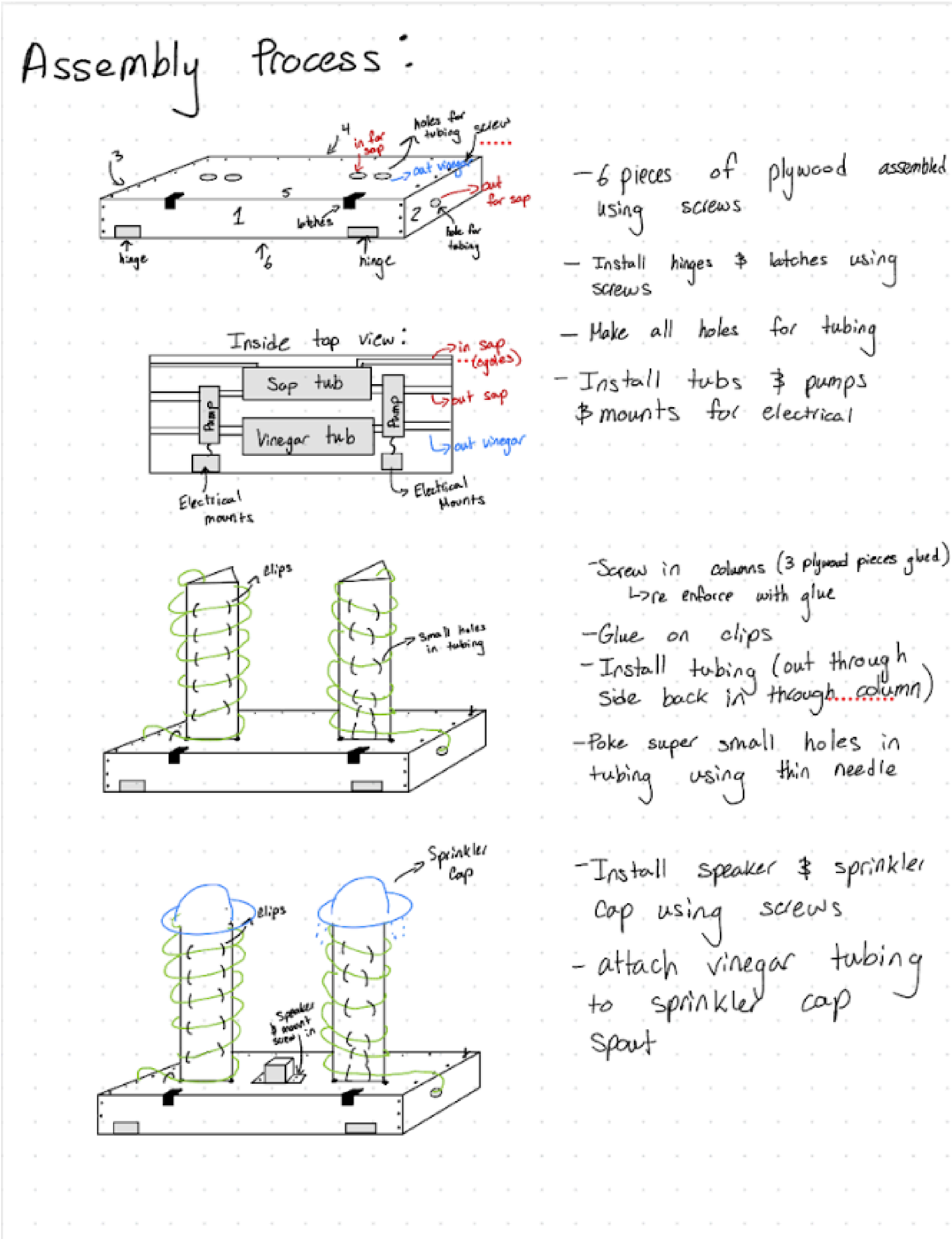
Step 3 – Vine Tubing Wrapping

- **Wrap tubing helically around poles**
- **Secure using printed clips**
- **Drill evenly spaced 1mm holes**
- **Test drip rates**
- **Install sprinkler caps to top of columns using screws**

Step 4 – Electrical Installation

- **Mount speaker centrally between poles**
- **Wire:**
  Battery → PDP → Cycle Timer → Pump
    - **PDP → Speaker]**

Sketch:


Assembly Process:
3
4
in for sap
holes for tubing
screw
out vinegar
5
1
2
out for sap
latches
hole for tubing
hinge
6
hinge
-6 pieces of plywood assembled using screws
- Install hinges & latches using screws
- Make all holes for tubing
- Install tubs & pumps & mounts for electrical
Inside top view:
in sap (cycles)
Sap tub
Pump
Pump
Vinegar tub
out sap
out vinegar
Electrical mounts
Electrical Mounts
clips
small holes in tubing
-Screw in columns (3 plywood pieces glued)
re enforce with glue
-Glue on clips
-Install tubing (out through side back in through column)
-Poke super small holes in tubing using thin needle
Sprinkler cap
clips
Speaker & mount screw in
-Install speaker & sprinkler cap using screws
- attach vinegar tubing to sprinkler cap spout